Panskura Banamali College (Autonomous)
B.Sc.(H) 5th Semester Examination 2022 (Online mode)
Subject : Computer Science Paper: DSE2P
Paper Name : Numerical Computing Lab
F.M. : 20 Time : 2 Hours

Instruction

[Make your answer script into PDF and MP3 file format and file name as like candidate_id-paper_code.pdf . For Example – candidate_id: 2020UG009967, paper_code : DSE2P , then file name will be :: 2020UG009967- DSE2P.pdf for question 'A' and 2020UG009967- DSE2P.mp3 for question 'B']

A. Answer any one question: **1 x12 = 12**

1. Write a program to Find the roots of the equation by secant/Regula−Falsi method in C language.
2. Write a program to find the roots of non-linear equation using Bisection method in C language.
3. Write a program to display the natural number upto 100 using array in C language.
4. Write a program to display the odd number upto 100 using array in C language.
5. Write a program to display the prime number upto 100 using array in C language.
6. Write a program to display the perfect square number upto 100 using array in C language.
7. Write a program to Find the solution of a system of nonlinear equation using Newton's method.
8. in C language.
9. Write a program to display the natural number upto 100 using array in C language.

Marks distribution:

a. Problem Definition - 1 Marks
b. Algorithm/Flow Chart/
Pseudo-code– 5 Marks
c. Program listing /coding- 3 Marks
d. Output/Result with discussion- 3 Marks

B. Answer any four questions: **4 x2 = 8**

[ONLINE Viva-Voce : Record the audio of your answers as like .mp3 format]

1. What is the purpose of having an additional array that carries along the values of an original array?
2. What is the purpose of a Transpose?
3. What form of numerical integration do you think is the most reliable?
4. Why do we bother with Newton Iterations when there are better way to solve things?
5. What is a Taylor Series Expansion?
6. How do you tie the iteration into the DO loop?
7. What is the difference between Newton's method and the Secant method?
8. What does Least Squares do?
9. When is Gauss Elimination Used?
10. How exactly do you use the cubic fit and what's its purpose?

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